
GOVERNMENT OF ANDHRA PRADESH

Chief Minister's Office - AP IAS Performance Tracking System

PERFORMANCE SCORE - 2 of 6

GSDP Score

An officer's contribution to the State's economic targets (Gross State Domestic Product) for 2025-26, attributed fairly and without double-counting across every post held.

Cells (district x sub-sector) - conserving day-share - efficiency x target-load factor

A plain-language note - every formula matches the live calculation.

1 What this score measures

GSDP (Gross State Domestic Product) is the total money value of all goods and services produced inside Andhra Pradesh in a year. The Government sets, for the year 2025-26, a TARGET and an ACHIEVEMENT for every combination of a district and an economic sub-sector. We call each such district-by-sub-sector combination a CELL. For example (Guntur district x Crops) is one cell, with its own target and achievement in rupees crore.

The GSDP Score measures how well the cells that an officer was responsible for met their targets - weighted by how long the officer held the relevant posts, and scaled up for officers who carried a larger total target. It is the only score with a formal CONSERVATION property: when every officer's attributed GSDP is added together, the grand total equals the State total (over the cells that have an officer). Nothing is invented; nothing is counted twice.

The data and the mapping

Sector / Sub-sector	the activity, e.g. Agricultural Sector > Crops, Livestock, Fisheries.
District	the geography; one cell = one district x one sub-sector.
Target 2025-26	the planned output value for that cell, in rupees crore.
Achievement 2025-26	the realised output value for that cell, in rupees crore.
gsdp_mapping table	the admin-maintained link from each sub-sector to the Department(s) and HoD(s) responsible for it (set on the GSDP-Mapping screen).

How an officer earns marks - the three kinds of post

An IAS officer rarely stays in one chair for the whole report period. They move through several posts, one after another, and occasionally hold two responsibilities at the same time. Every post record in the system carries a tag called the OFFICE STATUS. This tag is the single most important switch in the whole calculation, because it decides WHICH piece of real-world data is credited to that post. There are exactly three kinds:

- DISTRICT post - the officer heads a district (e.g. District Collector & Magistrate). The post is matched to that DISTRICT's data. The match key is the post's 'office location' field, which holds the district name.
- SECRETARIAT post - the officer heads a department at the State Secretariat (e.g. Principal Secretary, Special Chief Secretary). The post is matched to that DEPARTMENT's data, using the post's 'department' field.
- HoD post - the officer is the Head of a Department or a field organisation (e.g. Commissioner, Director, Director-General). The post is matched to that HoD's data, using the post's 'HoD' field.

A fourth possibility, 'Unmapped', occurs when a post has no usable office status (for example a training stint or a leave period). Unmapped posts carry no data and simply do not contribute to the score - they are neither rewarded nor penalised.

In plain words: the same officer can collect marks from a district while serving as Collector, from a department while serving as Secretary, and from a field office while serving as HoD. Each chair is judged only on the data belonging to that chair.

2 Which cells belong to a post

The office status decides the cells, but GSDP uses the mapping table to translate departments and HoDs into sub-sectors:

- DISTRICT post -> every sub-sector of that district. A Collector is treated as responsible for the whole district's economic output, so all of that district's cells belong to the post.
- SECRETARIAT post -> only the sub-sectors that the post's DEPARTMENT is mapped to (in gsdp_mapping), but across ALL districts. A Secretary's reach is state-wide for their subjects.
- HoD post -> only the sub-sectors that the post's HoD is mapped to, again across all districts.

For each cell the raw performance is simply the fraction of the target that was met:

CELL ACHIEVEMENT

$$\text{cell \% achieved} = \text{Achievement} / \text{Target} \times 100$$
Merging the same post and clamping to the GSDP year

Before counting days, an officer's records for the SAME post (identical name, location, department and HoD) are merged so over-lapping or touching stints become one continuous period and no calendar day is counted twice. Then only the portion of each period that falls inside the GSDP YEAR - 1 April 2025 to 31 March 2026, i.e. 365 days - is counted. A post held entirely before or after that window contributes zero.

3 Sharing each cell fairly - the conservation principle

A single cell can be the responsibility of more than one post at the same time. The clearest case: the (District-A x Crops) cell is part of the Collector of District A's responsibility AND part of the Secretary of Agriculture's responsibility. If the system credited the FULL target and achievement to BOTH, the State's total would be exaggerated - counted twice. To prevent this, each cell's value is SPLIT between the posts that cover it, in proportion to the days each post held it:

CONSERVING DAY-SHARE

$$\text{post's share} = \text{cell} \times \frac{\text{(this post's days on the cell)}}{\text{(total days of ALL posts on cell)}}$$

applied to BOTH the Target and the Achievement of the cell.

Why it is called 'conservation' (the proof in one line)

Add every post's share of one cell together. Each share is cell x (that post's days / total days); summing over all posts gives cell x (total days / total days) = cell x 1 = the cell's full value. So the parts always rebuild the whole. Therefore the sum over ALL officers of their attributed GSDP equals the sum of the cells - the State total - for every cell that has at least one officer.

In plain words: two officers responsible for the same target at the same time SPLIT it by their days, instead of each being handed the entire amount. The books balance to the State total.

Summing a post's shares over all its cells gives the post's PRO-RATA Target and Achievement. Summing those over the officer's posts gives the officer's AGGREGATED Target and AGGREGATED Achievement - the rupee figures shown on the dashboard and in the drill-down popup.

4 From achievement to the final score**Step A - Efficiency**

Efficiency is the day-weighted average of the posts' % achieved (using each post's in-year days as the weight). A post that achieved more of its target, and was held longer within the GSDP year, pulls the efficiency towards its result.

STEP A - EFFICIENCY

$$\text{Efficiency} = \frac{\text{SUM (post \% achieved \times FY-overlap days)}}{\text{SUM (FY-overlap days)}}$$

Step B - the target-load factor

The difficulty (load) factor - rewarding a heavier burden

Two officers may both be 90% efficient - but one cleared a tiny in-tray while the other ran a giant one; one managed a small grant while the other managed a huge one. Calling them equal would be unjust. So the efficiency is MULTIPLIED by a difficulty factor that grows with the size of the load the officer actually carried (the total GSDP target the officer carried).

DIFFICULTY / LOAD / VOLUME FACTOR

$$\text{difficulty} = \text{floor} + (1 - \text{floor}) \times W \quad (\text{floor} = 0.70)$$

$W = \log(1 + \text{load})$ placed on a 0..1 scale between the smallest and the largest log-load across all officers.

$W = 0$ for the smallest load, $W = 1$ for the largest.

load basis = the officer's AGGREGATED TARGET (rupees crore)

Why a logarithm?

A logarithm compresses big numbers so that equal MULTIPLES are equal steps. On a log scale the jump from 1 to 10 is the same size as 10 to 100, and 100 to 1000. We use it so that handling a much larger load is recognised as genuinely harder, WITHOUT letting a single enormous value swamp everyone else (which a plain, straight scale would allow).

In plain words: doing a small job perfectly can never out-rank doing a far bigger job equally well. The biggest load earns the full factor of 1.00; the smallest still keeps the floor (e.g. 0.70), so it is dampened but never reduced to zero.

Floor 0.70 and ceiling 1.00 are the defaults; the administrator can change both in the Score-Weights page. Setting floor = ceiling = 1.00 switches load-weighting off entirely, so the score becomes pure efficiency.

The final score

FINAL GSDP SCORE

$$\text{Load factor} = 0.70 + 0.30 \times W \quad (W = \text{log-scaled aggregated target})$$

$$\text{GSDP Score} = \text{Efficiency} \times \text{Load factor} \quad (0 - 100)$$

5 Worked example (the conservation in action)

WORKED EXAMPLE - GSDP

Officer held Collector-District-A (180 days) and Secretary-Agriculture (120 days), both within 2025-26. The cell (District-A x Crops) is covered by BOTH posts.

Cell (A x Crops): Target 100, Achievement 90.

Collector days on the cell = 180, Secretary days = 120, total = 300.

Collector's share : Target $100 \times 180 / 300 = 60$, Achievement $90 \times 180 / 300 = 54$.

Secretary's share : Target $100 \times 120 / 300 = 40$, Achievement $90 \times 120 / 300 = 36$.

Check: $60 + 40 = 100$ (target) and $54 + 36 = 90$ (achievement). Conserved.

After summing every cell of every post:

Aggregated Target = Rs.500 cr, Aggregated Achievement = Rs.430 cr.

Day-weighted Efficiency of post % achieved = 88.0

Large target -> $W = 0.80$ -> Load = $0.70 + 0.30 \times 0.80 = 0.94$

GSDP Score = $88.0 \times 0.94 = 82.7$

6 The statistics, explained

Day-weighted average

Posts are blended using their in-year days as weights, so a longer, better-achieving post counts for more than a brief stint. Posts outside the GSDP year drop out automatically (zero in-year days).

Conservation - why the totals always add up

The day-share split is engineered so the parts rebuild the whole (Section 4 proof). This is the statistical idea of a partition of unity: each cell's value is multiplied by fractions that sum to 1, so summing the parts returns the original. The dashboard header even shows the percentage of the State target that is attributed; any shortfall is simply cells that currently have no officer mapped to them, not a calculation error.

Logarithms for the load factor

The target-load factor uses log scaling so that carrying Rs.500 cr of target is recognised as meaningfully heavier than Rs.50 cr, fairly, without one very large officer dominating the whole comparison.

Pro-rata (proportional) arithmetic

Every rupee figure shown for an officer is pro-rata - sliced in proportion to days. This is the same idea as splitting a shared bill by the time each person was present, applied to targets and achievements.

7 Reading it on the dashboard & FAQ

The GSDP page shows each officer's Sectors Covered, Sub-sectors Covered, Aggregated Target, Aggregated Achievement, % Achieved, the GSDP Score and a state-wide Rank. Clicking the counts opens a drill-down: Sector > Sub-sector > Post > Period, every level showing the pro-rata target, achievement and %. The header shows the State target and the % attributed to officers.

Q. Two of us were in charge of the same district sub-sector. Do we both get full credit?

A. No - you share it by days. Together your shares add back to the cell's full value, so the State total is never exaggerated.

Q. I held a high-achieving post for only one month. Why is my score modest?

A. Day-weighting means a one-month stint contributes only its days to the average, and the target-load factor reflects the (small) total target you carried in the year.

Q. What is 'Aggregated Target'?

A. Your pro-rata share of all the cell targets you were responsible for, after the day-share split - i.e. the slice of the State target attributed to you.

Q. The header says only 92% is attributed. Is something wrong?

A. No - 8% of cells currently have no officer mapped to them (e.g. a vacant subject). Map them in the GSDP-Mapping screen and the figure rises.

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Glossary of terms

Cell

one district x sub-sector pair, with its own Target and Achievement.

Target / Achievement

planned vs realised output for a cell, in rupees crore.

gsdp_mapping

admin table linking sub-sectors to Departments and HoDs.

Pro-rata share

a cell's value sliced in proportion to a post's days on it.

Conservation

the day-share split whose parts always add back to the whole.

Aggregated Target/Achievement

an officer's summed pro-rata totals.

Efficiency

FY-day-weighted average of post % achieved.

Target-load factor

0.70-1.00 multiplier growing with the aggregated target.

FY 2025-26

1-Apr-2025 to 31-Mar-2026 (365 days) - the GSDP year.

Partition of unity

fractions that sum to 1, used to split a value without loss.

All weights, benchmarks and difficulty-factor floors/ceilings are stored in the score_config table and are configurable by the administrator in the Score-Weights page of the dashboard, so the system can be re-tuned without changing any code. Scores recompute automatically whenever the underlying data changes. Every formula in this note matches the live calculation exactly.